Snaps from space

Hubble, the world's first large space telescope, was launched in 1990 and orbits Earth at an altitude of 340 miles (547 km). Free of the distorting effects of Earth's atmosphere, the images it collects from light gathered in by its mirror are between five and 20 times sharper than those obtained from the ground. This has enabled scientists to detect phenomenally distant nebulae (clouds of hot gas and dust from which stars are made) and galaxies, some as far as 13.4 billion light years away. Hubble has made more than 1.2 million observations since its launch. It sends an average of 140 gigabytes (GB) of data back to Earth each week yet requires just 2,100 watts (W) of power, little more than an electric kettle.

The Carina Nebula's swirling clouds of gas, dust, and young, bright stars is highlighted in this composite of 44 images taken by the Hubble. This nebula lies around 7,500 light years from Earth.